

Nonlinear Productivity Dynamics

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Productivity differences across firms are widely documented.

- ▶ Firm-level productivity is **dispersed** in the cross-section and **persistent** but not permanent over time.

Two main questions:

- ▶ *Which productivity drives these differences?*
Technical efficiency (TFPQ) [exogenous] or ability to generate \$ (TFPR = TFPQ + Price) [endogenous]?
- ▶ *How do the dynamics in TFPQ and TFPR compare?*
Is the persistence of productivity histories coming from technology or demand?

Two main contributions:

- ▶ *Methodological*: iterative framework to separately identify TFPQ and Demand.
- ▶ *Empirical*: document pervasive deviations from linear AR(1) in the cross-section and dynamics, in a sample of PT manufacturers.

Framework

TFPQ vs TFPR: definitions

TFPQ (Z^Q): *how well a firm turns inputs into units of output* (technology)

$$Q_{it} = Z_{it}^Q K_{it}^{\beta_K} L_{it}^{\beta_L}$$

Z_{it}^Q = **TFPQ**: Exogenous physical efficiency (Hicks-neutral)

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TFPR (Z^R): *how well a firm turns inputs into revenue (technology + demand).*

$$Z_{it}^R = P_{it} \cdot Z_{it}^Q$$

$\Rightarrow P_{it}$ brings **market structure** (market power, strategic interactions, brand appeal,..) into the standard "revenue productivity" measures.

$$\text{Revenue} = P_{it} Q_{it} = P_{it} Z_{it}^Q K_{it}^{\alpha_K} L_{it}^{\alpha_L} \Leftrightarrow \ln(\text{Revenue}) = \underbrace{\ln(P_{it} Z_{it}^Q)}_{\text{common productivity measure}} + \alpha_K \ln K_{it} + \alpha_L \ln L_{it}$$

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Example: Two bakeries, identical inputs

	Output	Price	Revenue	Implication
Bakery A	100 loaves	€2	€200	High TFPQ, Low TFPR
Bakery B	80 loaves	€3	€240	Low TFPQ, High TFPR

TFPQ vs TFPR: a basic framework

1. **Production:** TFPQ dynamics need not be linear-Gaussian.

$$z_{it}^q = g(z_{i,t-1}^q, \xi_{it}^q)$$

$g(\cdot)$ potentially nonlinear with non-separable non-Gaussian innovations (ξ_{it}^q).

⇒ persistence $\partial z_{i,t}^q / \partial z_{i,t-1}^q = \rho(z_{i,t-1}^q, \xi_{it}^q)$ varies;

⇒ heteroskedasticity, state-dependent high-order moments.

2. **Demand:** generic monopolistic competition with non-CES

$$Q_{it} = f(P_{it}, MS_{it}, \underbrace{D_{it}}_{\text{demand shifters}})$$

⇒ variable markups, allowing strategic interaction via market-share MS_{it} (Atkeson–Burstein).

3. **Revenue:** (1) + (2) ⇒ rich pricing behaviour

$$P_{it} = h(z_{it}^q, D_{it}, MS_{it}, K_{it})$$

⇒ local pass-through rate TFPQ → Prices: $\psi \equiv -\partial \log h / \partial z^q$;

⇒ **TFPQ** ≠ **TFPR** in the cross-section and in dynamics.

TFPQ vs TFPR: why it matters

1. **Firm-level responses:** non-linear persistence of TFPQ and state-dependent volatility change policy functions.
 - ▶ Memory-wiping shocks: extreme shocks can erase history, making the past less relevant for future decisions.
 - ▶ Pricing decisions: dampen or amplify TFPQ shocks $\Rightarrow$ TFPR dynamics capture menu costs, market power, strategic interactions,...
2. **Aggregate level:**
 - ▶ TFPR not a monotone function of TFPQ $\Rightarrow$ careful interpretation of dispersion as measure of misallocation.

Preview of empirical findings

1. TFPQ is non-Gaussian and dynamically nonlinear.

- ▶ *Heavy tails, negative skewness; dispersion and skewness vary across the distribution.*
- ▶ *Persistence depends on level $\times$ shock: large extreme shocks “wipe memory” at the top and bottom of the productivity distribution.*

2. TFPR inherits and amplifies these features.

- ▶ *Greater dispersion, stronger skewness, fatter tails than TFPQ.*
- ▶ *TFPR is more persistent than TFPQ — pricing/markup behaviour smooths productivity shocks.*

3. Similar-TFPR firms mask different technical efficiencies.

- ▶ *Only 50% of firms in the top quantile of TFPR are also in the top quantile of TFPQ.*

4. TFPQ pass-through to markups is asymmetric.

- ▶ *Below-median TFPQ firms transmit productivity into markups; above-median firms barely do.*

Roadmap

Literature

Related Literature

1. Production Function Estimation: [Olley and Pakes (1996), Levinsohn and Petrin (2003), Akerberg et al. (2015), Eslava et al. (2024) and De Loecker (multiple)]

- ▶ **Our contribution:** (1) incorporate demand-side factors in production function estimation; (2) allow for nonlinear dynamics in productivity (go beyond AR1).

2. Non-Gaussianity & Non-Linearity in Firm Dynamics: [Jaimovich et al.(2025), Guvenen et al. (2023), Fella et al. (2022), Ruiz-Garcia (2021), Aguirre (2023)]

- ▶ Some micro-evidence of nonlinearities in revenue function residuals (TFPR proxy).
- ▶ **Our contribution:** breakdown non-linear dynamics in **TFPQ** (for the first time) vs TFPR.

3. TFPQ vs. TFPR: [Foster et al. (2008, 2016), Eslava et al. (2023), De Vera(2022)]

- ▶ Determinants of firm growth: technical efficiency vs demand advantages.
- ▶ **Our contribution:** *jointly* estimate demand-side advantages, TFPQ, and TFPR, in a theory-consistent fashion.

Estimation Strategy

Estimating the gross production function

Why not simply run an OLS on the gross-output production function?

$$\ln(Y_{it}) = \ln z_{it}^Q + \alpha_K \ln K_{it} + \alpha_L \ln L_{it} + \alpha_M \ln M_{it} + \epsilon_{it}$$

1. Productivity unobserved by the econometrician $\implies$ part of the residual.
2. But known to firm $\implies$ input choices of M are correlated with residual; K, L predetermined.

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Common antidote: Control function approach (ACF, 2015) — use materials demand as a proxy for productivity.

- *Why?* Materials are chosen **after** observing productivity: $m_{i,t} = m(z_{i,t}^q, \mathbf{X})$.
 $\Rightarrow$ Materials demand is monotone in z^Q , *conditional on the information set* $\mathbf{X}$
 $\Rightarrow$ it can be inverted to control for the unobservable: $z_{i,t}^q = m^{-1}(m_{i,t}, \mathbf{X})$.

But the standard ACF approach still leaves two challenges unresolved.

Remaining challenges with the standard approach

1. Demand contaminates productivity estimates

- 1.1 When deflated revenues (\$) instead of quantities are used as Y any 'productivity' measure conflates physical efficiency with demand (prices).
- 1.2 Even when using quantities, demand aspects (e.g. market share, brand appeal) can affect input choices, contaminating the proxy if not featured in **X**.

2. Nonlinear productivity dynamics are ruled out

Standard ACF imposes an AR(1) law of motion — potentially misspecified.

Roadmap to address the identification challenges

1. Demand contaminates productivity estimates

1.1 Use quantities of output **produced** (not sold) measured in consistent units as Y .

1.2 Include demand shifters in $\mathbf{X}$, both observable and latent (*demand advantages*).

- ▶ Back out *demand advantages* from the **demand system** and feed them into $\mathbf{X}$.
- ▶ Use TFPQ as supply-side instrument for prices (Foster et al. 2008)

2. Nonlinear productivity dynamics are ruled out

- ▶ Productivity allowed to follow a very flexible high-order polynomial $g(z_{i,t-1}^q, \xi_{it}^q)$.
*Instead of ex-ante imposing linearity (typical approach). Nests AR(1).
Robust to asymmetric and heavy-tailed shocks*

Roadmap to address the identification challenges

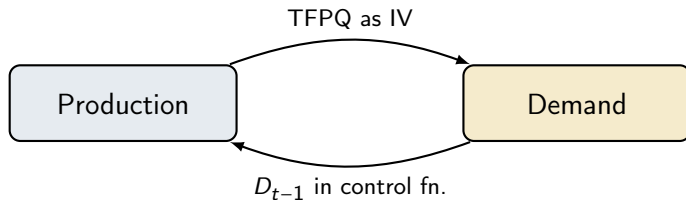
Two interdependent blocks:

1. Production function $\rightarrow$ TFPQ

- ▶ Control function approach (ACF)

2. Demand system $\rightarrow$ Demand advantages

- ▶ Instrument prices with TFPQ (Foster et al. 2008) in a richer demand setup estimated via GMM à la De Vera (2024).



Solution: Iterative algorithm that alternates between blocks.

▶ Iterative Algorithm

Assumptions

- A1 Information Set:** $\mathcal{I}_{it}$ includes $\{z_{i\tau}^q\}_{\tau=0}^t$ and $\{D_{i\tau}\}_{\tau=0}^{t-1}$, but not future shocks.
- A2a Markov Productivity:** $p(z_{it+1}^q | \mathcal{I}_{it}) = p(z_{it+1}^q | z_{it}^q)$ (Unobserved by the econometrician)
- A2b Markov Demand:** $p(D_{it+1} | \mathcal{I}_{it}) = p(D_{it+1} | D_{it})$ (Unobserved by the econometrician)
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- A3 Timing:**
- Capital k_{it} and Labour l_{it} : determined at $t-1$
 - Productivity z_{it}^q : realized at **start of t**
 - Materials chosen at t : $m_{i,t} = m(z_{i,t}^q, \mathbf{X})$
 - Demand D_{it} : realized at **end of t**

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- A4 Materials demand function** takes as arguments $\mathbf{X} \in \{k_{it}, l_{it}, D_{i,t-1}, MS_{i,t}\}$
- A5 Strict Monotonicity:** $m(\cdot)$ strictly monotonic in z_{it}^q conditional on $\mathbf{X}$

Focus on A2a: Markov Productivity

A first-order Markov productivity z_{it}^q admits parametrizations of increasing generality.
Three illustrative DGPs (shock ξ_{it}^q , generic distribution) - all nested in our estimation.

[DGP1] Linear (classical) AR(1).

Linear, homoskedastic.

$$z_{it}^q = \rho z_{i,t-1}^q + \xi_{it}^q; \quad \rho(z_{i,t-1}^q, \xi_{it}^q) = \rho.$$

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[DGP2] Nonlinear AR(1) with state-dependent variance *Nonlinear drift + heteroskedasticity*

$$z_{it}^q = \rho_0 z_{i,t-1}^q + \rho_1 (z_{i,t-1}^q)^2 + (1 + \gamma z_{i,t-1}^q) \xi_{it}^q$$
$$\rho(z_{i,t-1}^q, \xi_{it}^q) = \rho_0 + 2\rho_1 z_{i,t-1}^q + \gamma \xi_{it}^q.$$

[DGP3] Hermite polynomial (à la ABB). *Shock quantile-dependent polynomial coefficients.*

$$z_{it}^q = a(\xi_{it}^q) + b_1(\xi_{it}^q) z_{i,t-1}^q + b_2(\xi_{it}^q) (z_{i,t-1}^q)^2$$
$$\rho(z_{i,t-1}^q, \xi_{it}^q) = b_1(\xi_{it}^q) + 2 b_2(\xi_{it}^q) z_{i,t-1}^q.$$

⇒ **Memory-wiping:** [2] and [3] allow current shocks to change the persistence of past productivity.

Algorithm - the Production Function Block

First stage — Control-function projection.

- ▶ Project log production (in units) q_{it} on a flexible high-order polynomial $\Phi_t(l, k, m, D_{t-1}, MS)$, including the inverted material-demand $m(\cdot)$.
- ▶ Recover $\hat{\Phi}_{it}$, cleansed of m.e.. Input elasticities $\beta = (\beta^L, \beta^K, \beta^M)$ *not yet identified*.

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Second stage — GMM on the productivity innovation.

For a candidate β :

1. Trial productivity: $\hat{z}_{it}^q(\beta) = \hat{\Phi}_{ist} - \beta^L l_{ist} - \beta^K k_{ist} - \beta^M m_{ist}$.

2. Inner projection on a polynomial in the lag (conditional mean):

$$\hat{z}_{it}^q = \tilde{g}\left(\hat{z}_{i,t-1}^q\right) + \tilde{\xi}_{it}(\beta), \quad \mathbb{E}\left[\tilde{\xi}_{it} \mid \hat{z}_{i,t-1}^q\right] = 0 \text{ (by construction).}$$

3. GMM moment condition:

$$\mathbb{E}[\tilde{\xi}_{it}(\beta) \cdot Z_{it}] = 0$$

$$Z_{it} = (k_{it}, l_{i,t-1}, m_{i,t-1}, \dots)$$

$\tilde{\xi}_{it}$ is *not* the TFPQ shock ξ_{it}^q — and that's fine. ▶ why?

Identification of β requires only that $\tilde{g}(\cdot)$ span the true conditional mean. Distributional or heteroskedasticity features of ξ^q propagate into higher moments of $\tilde{\xi}$ — they affect efficiency, not consistency.

Algorithm - the Demand Function Block

Demand equation (Atkeson–Burstein, log-linear variable-markup).

$$q_{it}^{\text{sold}} = \alpha_p p_{it} + \alpha_{pms} (p_{it} \cdot \log MS_{it}) + D_{it}$$

- ▶ q^{sold} is log *quantity sold* (not produced) — the schedule the consumer actually clears.
- ▶ D_{it} : firm-specific demand advantage (brand, location, quality, ...).
 - ▶ Follows the process: $D_{it} = \rho_0 + \rho_1 D_{i,t-1} + \rho_2 D_{i,t-1}^2 + \xi_{it}^d$

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- ▶ **Identification — GMM with supply-side instrument.** p_{it} is endogenous (clears the market after D realizes). TFPQ $\hat{z}_{it}^q$ shifts the supply schedule but is orthogonal to the demand innovation ξ_{it}^d ; combined with lagged prices, quantities, and capital.

$$\mathbb{E}[\xi_{it}^d \cdot Z_{it}] = 0$$

Output. $(\hat{\alpha}_p, \hat{\alpha}_{pms}, \hat{\rho}_0, \hat{\rho}_1, \hat{\rho}_2)$ and recovered $\hat{D}_{it}$, which feeds back into the production-side control function.

Algorithm - does it work?

Before taking it to the data, 2 questions rise about our proposed iterative algorithm:

1. Does it recover the truth?

- ▶ Simulate (**Monte Carlo**) data from a menu of productivity processes (DGP1, DGP2, DGP3).
- ▶ Compare estimates to known DGP parameters.

2. How precise are the empirical estimates?

- ▶ **Panel Block Bootstrap**: resample firms with replacement.
- ▶ Re-run the full iterative procedure on each replication
- ▶ Standard errors for $\beta^L, \beta^M, \beta^K, \eta$.

1. The algorithm recovers the true elasticities.

True parameters: $\beta^L = 0.3$, $\beta^M = 0.5$, $\beta^K = 0.2$, $\eta = -2.5$.

Sample size: $N_{\text{firms}} = 2,500$ per replication, $T = 10$ years, one sector, balanced panel.

Table 1: Monte Carlo: Production Function Estimates Across Selected DGPs

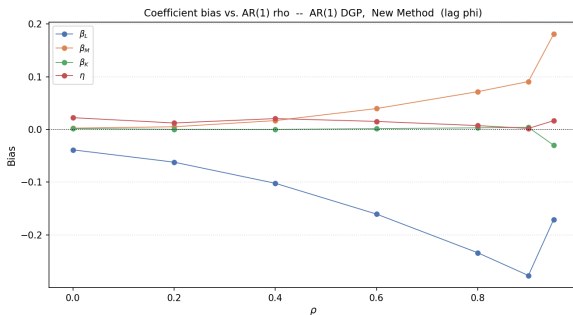
DGP	Stage	β^L (0.30)	β^M (0.50)	β^K (0.20)	η (-2.50)
DGP1 (standard AR(1))	Iter. 1	0.216 (0.012)	0.620 (0.025)	0.215 (0.007)	—
	Final	0.300 (0.010)	0.497 (0.008)	0.203 (0.009)	-2.44
DGP2 (Nonlinear AR)	Iter. 1	0.282 (0.006)	0.621 (0.020)	0.204 (0.011)	—
	Final	0.308 (0.008)	0.474 (0.010)	0.220 (0.009)	-2.53
DGP3 (Hermite Polynomial)	Iter. 1	0.211 (0.018)	0.694 (0.039)	0.289 (0.022)	—
	Final	0.270 (0.014)	0.497 (0.016)	0.248 (0.037)	-2.17

⇒ Iter. 1 is biased; final iterative estimator is on target across all DGPs.

2. If demand and prod. shocks are (too) correlated, the identification fails.

A2c: $\xi_{it}^q \perp \xi_{it}^D$.

Violation tested: $\text{cov}(\xi_{t-1}^D, \xi_t^q) = \sigma^{\text{cross}} \in \{0, 0.2, \dots, 0.95\}$ — lagged demand shocks predicting current TFPQ shocks (the relevant case, since D_{t-1} is in the materials proxy).



Bias of $\hat{\beta}^L, \hat{\beta}^M, \hat{\beta}^K, \hat{\eta}$ under AR(1) DGP, our iterative method.

$\Rightarrow$ All four parameters are essentially unbiased at the identifying assumption ($\sigma^{\text{cross}} = 0$); bias grows smoothly in β^L, β^M as the assumption is violated, while β^K and η remain robust.

Application: Portuguese Manufacturing Firms

Data sources

We merge 3 microdatasets on Portuguese manufacturing firms, 2010-2022:

- ▶ **Balance sheet and income statement data:** *BS* (Balance Sheet). **Near population coverage, yearly.**
- ▶ **Firm characteristics:** sector, firm age, entry/exit from *Business Register* (BR). **Population coverage, yearly.**
- ▶ **Quantities and prices (produced and sold):** *PRODCOM*, i.e. product-level information on output, for manufacturing firms. **Survey data, yearly.**

▶ Data Cleaning

Data obtained via CompNet-MDI.

The analysis could be extended to other countries.

Industries in focus

The level of aggregation in the production function estimation is a tradeoff:
product homogeneity vs. sample size.

⇒ Here: *industry* refers to the 7-digit product code level in PT PRODCOM.

- Firms are coded based on main product (i.e. at least 70% of sales in t).

Product Code	Unit	Obs	Firms	Description
1051405	kg	702	133	Cheese products
1071110	kg	1828	418	Bread products
1391191	kg	729	127	Knitted and crocheted fabrics
1414300	pc	993	179	Underwear garments
1520135	pc	6166	760	Leather footwear
2512105	pc	8882	1323	Metal doors, windows and frames
2573507	pc	3329	449	Metal tools, locks and hardware
3102100	pc	1918	240	Wooden kitchen furniture

Table 2: Industries in sample

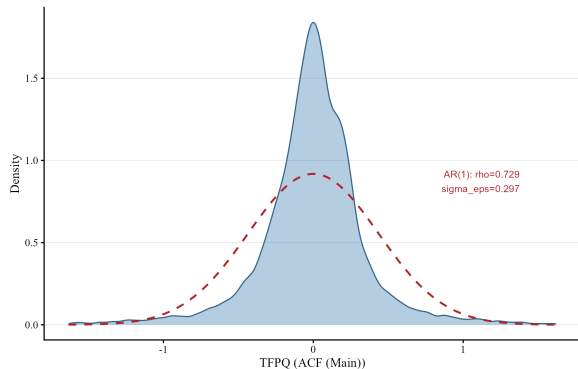
Descriptive statistics

- ▶ Sample of established medium-sized manufacturing firms (25,283 observations)
 - ▶ Descriptive Statistics Portuguese Firms
- ▶ Average firm is observed for 6 years. Unbalanced panel. Entry/exit into sample **unrelated** with entry/exit into business.
- ▶ Majority of firms have 1 single plant

Table 3: Summary Statistics for Selected Variables

	P ₂₅	P ₅₀	P ₇₅	Mean
Year	2014	2017	2019	2016.76
Age	9	17	26	18.64
Headcount	5.00	10.00	26.50	24.57
Turnover (in Thd.)	188.5	445.26	1354.74	1673.01
Capital (in Thd.)	20.86	90.07	352.74	552.82

1. TFPQ: unconditional distribution is heavy-tailed



Key observations:

- ▶ Fat tails (more extremes than AR1-Gaussian)
- ▶ Negative skewness (longer left tail)
- ▶ More 'peaked'

2. TFPQ: conditional distributions reveal state-dependent uncertainty

Break down TFPQ into quantiles and study $z_t^q | z_{t-1}^q$:

- ▶ Bottom-TFPQ firms are more exposed to left-tail shocks and top-TFPQ firms to right-tail shocks.

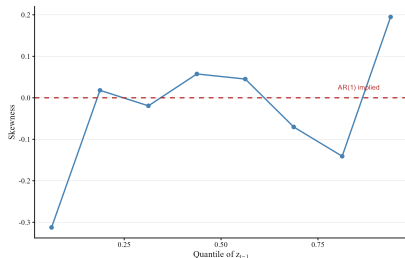


Figure 1: Conditional skewness

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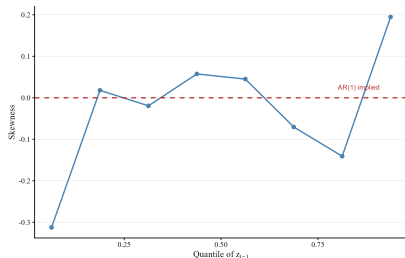


Figure 1: Conditional skewness

- ▶ Firms at the extremes face more uncertainty [**u-shaped dispersion**].

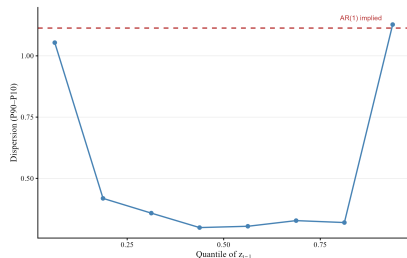


Figure 2: Conditional dispersion

⇒ Low vs high TFPQ firms face substantially different stochastic processes!

3. TFPQ: persistence is nonlinear (no constant ρ)

Memory-wiping (low ρ) occurs at the extremes

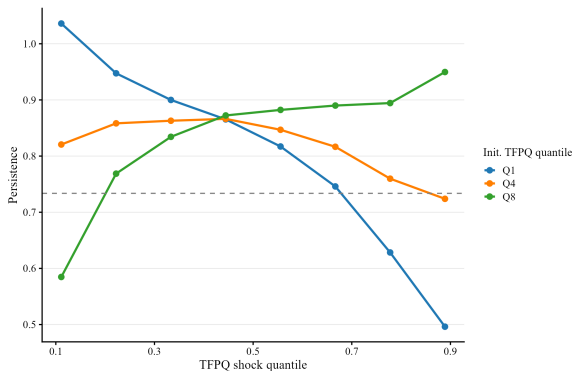


Figure 3: Persistence as function of TFPQ shock (quantile), for different inherited TFPQ levels.

For low-TFPQ firms (Q1):

- ▶ Large *positive* shocks (high τ)
- ▶ $\Rightarrow$ Low persistence
- ▶ $\Rightarrow$ “Escape” low productivity

For high-TFPQ firms (Q8):

- ▶ Large *negative* shocks (low τ)
- ▶ $\Rightarrow$ Low persistence
- ▶ $\Rightarrow$ Top firm status erased

Mid-TFPQ firms: Stable, high persistence regardless of shock ($\approx$ AR(1))

3. TFPQ: persistence is nonlinear (in 3D)

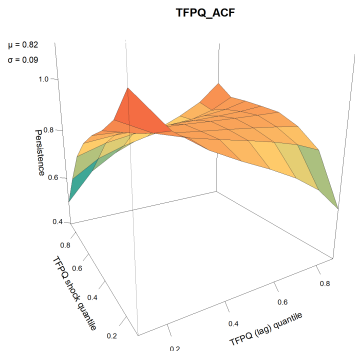


Figure 4: Estimated persistence of TFPQ shocks

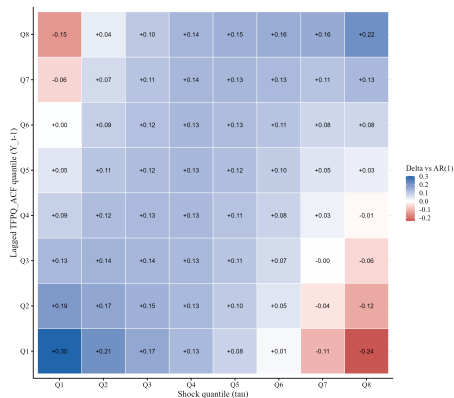


Figure 5: Difference to AR(1) benchmark

► Estimated AR(1) moment

► Heterogeneity Age

► Heterogeneity Size

4. Absent demand controls in estimation, TFPQ rankings are misleading.

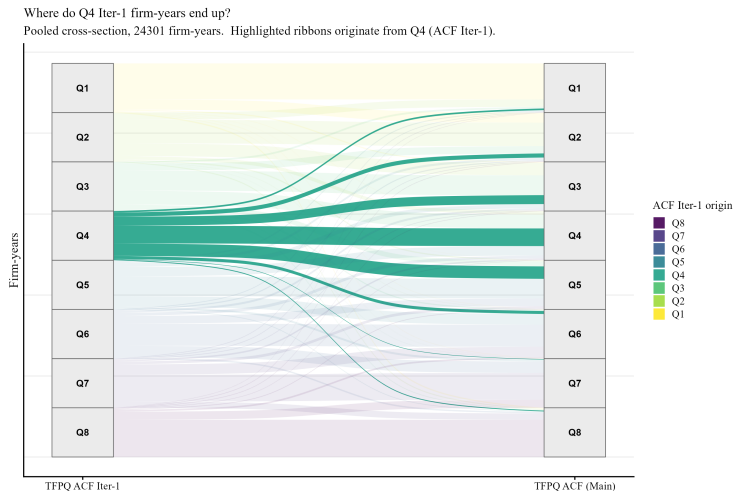


Figure 6: Changes in TFPQ ranking after controlling for unobserved demand.

From TFPQ to TFPR

1. TFPR distribution: further deviations from linear-Gaussian benchmark

Remember:

$$z_{it}^r = z_{it}^q + p_{it}$$

We find that TFPR **inherits and amplifies** many of the 'nonstandard' features of TFPQ:

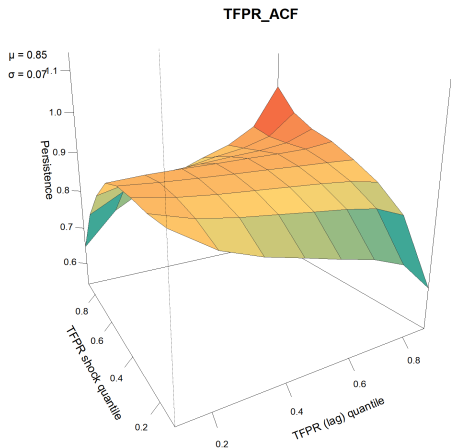
- ▶ Higher dispersion
- ▶ Positive skewness
- ▶ Heavier tails

	TFPQ	TFPR
Std. Dev.	0.4	0.65
Dispersion	0.72	1.14
Skewness	-0.03	0.14
Kurtosis	1.73	1.94

▶ Cond. Moments

▶ Distribution overlay

2. TFPR is more persistent than TFPQ throughout



Average persistence:

► TFPQ: $\bar{\rho} = 0.73$

► TFPR: $\bar{\rho} = 0.82$

Why higher?

► Prices are persistent

3. Do firms rank similarly in TFPQ and TFPR?

Overall correlation: 0.54

But rank mobility varies:

- ▶ **Extremes:** High concordance
 - ▶ 65% of bottom TFPQ → bottom TFPR
 - ▶ 49% of top TFPQ → top TFPR
- ▶ **Middle:** Substantial mobility
 - ▶ Median TFPQ spreads across all TFPR deciles
 - ▶ Prices bring average-productive firms to the full spectrum of TFPR.

	z_1^r	z_2^r	z_4^r	z_6^r	z_8^r
z_1^q	0.65	0.17	0.01	0.01	0.11
z_2^q	0.14	0.46	0.05	0.02	0.07
z_4^q	0.03	0.08	0.32	0.08	0.04
z_6^q	0.02	0.05	0.11	0.30	0.08
z_8^q	0.04	0.01	0.01	0.06	0.49

TFPQ to TFPR transition

3. Do firms rank similarly in TFPQ and TFPR?

Where do top-TFPR (Q8) firm-years come from?

Pooled cross-section, 24547 firm-years. Highlighted ribbons end at TFPR Q8.

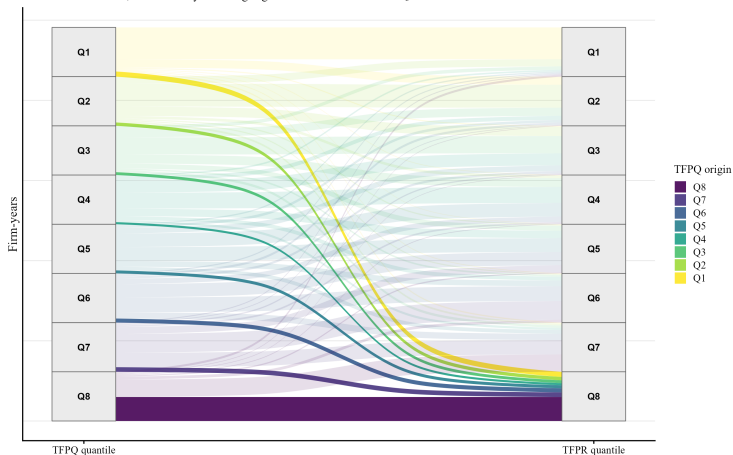


Figure 7: Flow chart: where do top-TFPR (Q8) firms come from in terms of TFPQ?

4. Markups move asymmetrically with TFPQ

$$z_{it}^r = \underbrace{z_{it}^q}_{\text{TFPQ}} + \underbrace{p_{it}}_{\text{Price}} = z_{it}^q + \underbrace{\mu_{it}}_{\text{Markup}} + \underbrace{mc_{it}}_{\text{Marginal cost}}$$

► Markup computation

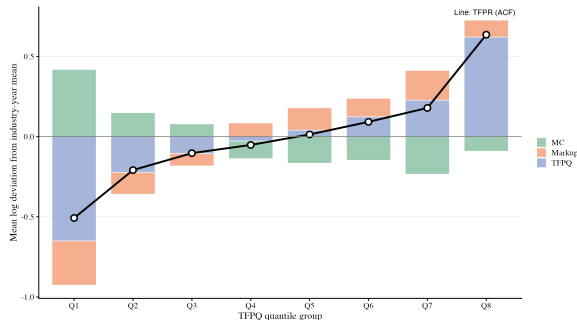


Figure 8: Average markup, marginal cost, TFPQ and TFPR - by TFPQ quantile.

Why this matters?

1. Pass-through from TFPQ to prices and markups

Our method cleanses TFPQ of demand confounders, which has important implications vis-à-vis the traditional ACF AR(1) approach:

- ▶ Less persistence in TFPQ over time
- ▶ Higher pass-through from TFPQ to TFPR
- ▶ Less reaction in prices
- ▶ Stronger reaction in markups, especially for high-markup firms

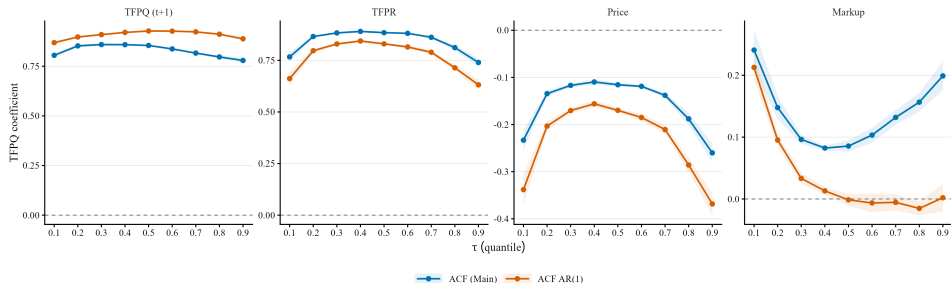


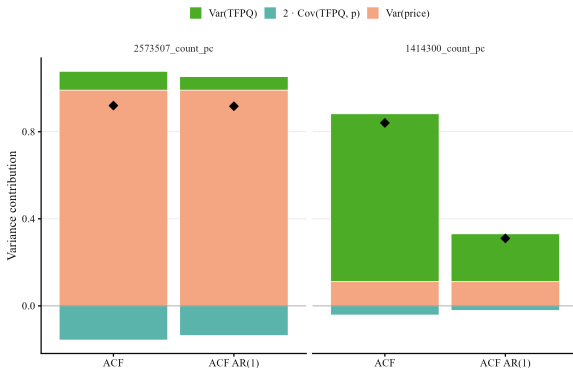
Figure 9: Quantile regression by method: ACF (Main) vs. ACF AR(1)

2. Productivity dispersion ('misallocation') picture changes

$$\text{Var}(z^r) = \text{Var}(z^q) + \text{Var}(p) + 2 \text{Cov}(z^q, p)$$

Var(log TFP_r) decomposition: ACF vs. ACF AR(1)

Two selected sectors. Diamond marks Var(log TFP_r). 2*Cov can be negative; bars dip below 0.



Left pair: *metal tools*. Right pair: *apparel/clothes*.

Proposed methodology bites where demand advantages are more salient (greater differentiation).

- ▶ **Metal tools** (homogeneous): controlling for demand moves TFP_q estimates by little.
- ▶ **Apparel** (differentiated): proposed methodology uncovers much higher $\text{Var}(z^q)$.

Conclusion

Conclusion

Main Findings:

- ▶ TFPQ dynamics are **nonlinear**: conditional moments depend on past productivity
- ▶ **Memory-wiping** at the extremes: large shocks can erase productivity history
- ▶ TFPR inherits these nonlinearities but is **more persistent** due to prices

Why It Matters:

- ▶ For firms: being subject to "memory-wiping" shocks should impact policy functions.
- ▶ For aggregates: high-TFPR firms can actually be inefficient (low TFPQ). Different welfare implications of policies targeting TFPQ vs. TFPR.

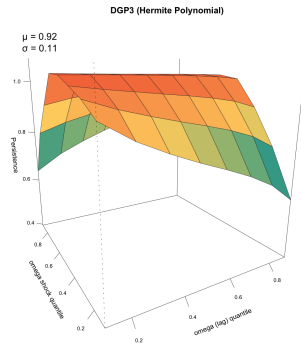
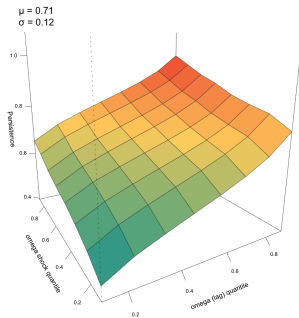
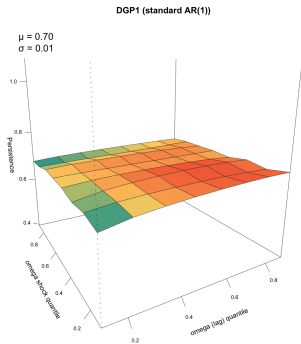
Methodology:

- ▶ Joint estimation of production and demand advantages via iterative algorithm
- ▶ Featuring flexible functional allowing nonlinearities in dynamics.

Thank you.

MC: Implied persistence surfaces across DGPs

Persistence surfaces across DGPs



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Convergence criteria of the iterative algorithm

Fixed-point loop. Alternate two blocks until joint convergence:

- ▶ **GMM step:** update $(\eta, \rho_0, \rho_1, \rho_2)$ from demand-side moments, holding TFPQ fixed.
- ▶ **ACF step:** update $(\beta^L, \beta^M, \beta^K)$ via production-function GMM, holding η fixed; recompute demand shocks $\hat{\phi}_{it}$.

Damping on η : $\eta \leftarrow 0.5\eta_{\text{new}} + 0.5\eta_{\text{old}}$.

Stopping rule. Convergence declared when *all* hold simultaneously:

- ▶ $\max\{|\Delta\eta|, |\Delta\beta^L|, |\Delta\beta^M|, |\Delta\beta^K|\} < 0.03$
- ▶ $\|(\Delta\eta, \Delta\beta^L, \Delta\beta^M, \Delta\beta^K)\|_2 < 0.03$

Cap: max. iterations = 100.

Multiproduct firms

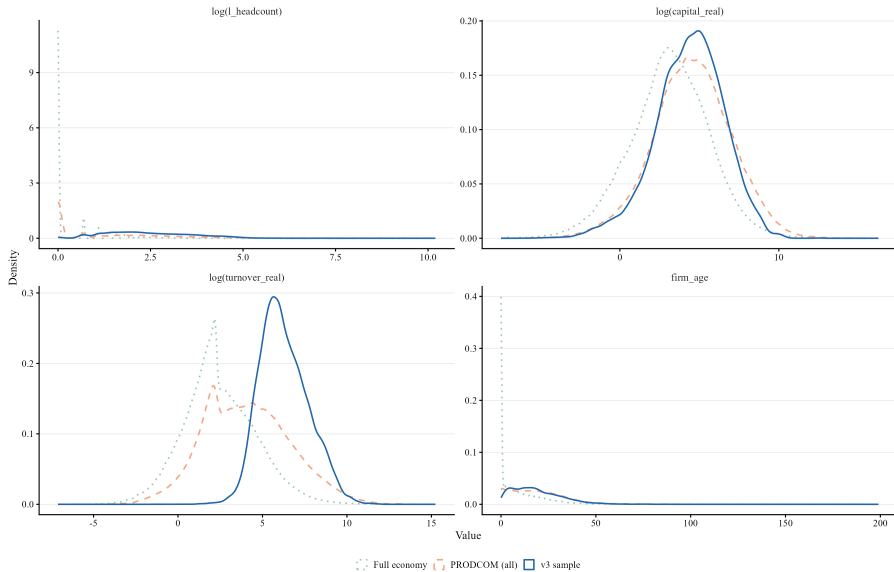
For now handled very basically, aggregated based on unit of measurement similarity.
An example:

year	firmid	product code	description	unit	q	$q \times p$
2015	501112773	152011351	Men's town footwear with leather uppers	pa	35909	1000567
2015	501112773	152011352	Women's town footwear with leather uppers	pa	39384	1058562
2015	501112773	152011353	Children's town footwear with leather uppers	pa	7349	194643
2015	5011127	--	--	pa	82642	2253772

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Descriptive statistics

Sample selection: firm-level distributions



Data Cleaning Pipeline

1. Harmonize units of measurement
2. Drop missing values in key variables
3. Keep privately owned firms only
4. Exclude overseas territories
5. Minimum thresholds: assets, VA, employees
6. Require 3-year consecutive observations per firm
7. Require sufficient observations per industry (industry = product $\times$ unit)
8. Final sample: 25,283 observations

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Iterative Algorithm

Step 0: Initialize with standard ACF (no demand controls)

Step 1: *Demand block*

- ▶ Use current TFPQ as instrument for prices
- ▶ Recover $D_{it} = y_{it} - \hat{\alpha}_p p_{it}$

Step 2: *Production block*

- ▶ Include lag of D_{it} in control function
- ▶ Re-estimate production function (quantile GMM)
- ▶ Recover TFPQ

Step 3: Iterate Steps 1–2 until convergence on estimated parameters.

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Moments

Quantile-based measures of dispersion, skewness, and kurtosis:

$$\text{Dispersion} = P_{90} - P_{10},$$

$$\text{Skewness} = \frac{(P_{90} - P_{50}) - (P_{50} - P_{10})}{P_{90} - P_{10}},$$

$$\text{Kurtosis} = \frac{P_{92} - P_8}{P_{83} - P_{17}}.$$

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An alternative: a general first-order Markov process

Quantile-based panel data method. A stochastic process modeled through a series of **conditional quantile functions (CQF)**.

1. Allow lagged TFPQ and current shock to nonlinearly interact:

- ▶ $Q_z(z_{i,t-1}^q, \tau)$ yields the current TFPQ level for a firm with some past TFPQ ($z_{i,t-1}^q$) when facing shock of rank τ in t .

$$z_{it}^q = Q_z(z_{i,t-1}^q, u_{it}), \quad (u_{it} | z_{i,t-1}^q, z_{i,t-2}^q, \dots) \sim U(0,1), \quad t > 1.$$

- ▶ For most of analysis, we consider 8 quantile points, i.e. $\tau \in \{0.11, \dots, 0.88\}$.
- ▶ In practice, we estimate a dictionary of low-order Hermite polynomials at each τ .
- ▶ Nests AR(1) if linearity + Gaussian shocks assumed:

$$z_{it}^q = \rho z_{it-1}^q + \Phi_t^{-1}(u_{it}; \sigma)$$

(where Φ_t^{-1} is the inverse CDF of Normal with mean zero and sd σ).

Why $\tilde{\xi}$ differs from ξ^q — and β is still identified

What $\tilde{\xi}$ is. The inner projection on a high-order polynomial in the lag returns the *conditional-mean residual*; at the true β_0 ,

$$\tilde{\xi}_{it} = \hat{z}_{it}^q - \tilde{g}(\hat{z}_{i,t-1}^q) = \underbrace{[\mu(\hat{z}_{i,t-1}^q) - \tilde{g}(\hat{z}_{i,t-1}^q)]}_{\text{misspec gap}} + \underbrace{[\hat{z}_{it}^q - \mu(\hat{z}_{i,t-1}^q)]}_{\text{true innovation, } \perp \sigma(\hat{z}_{i,t-1}^q)},$$

with $\mu(\cdot) \equiv \mathbb{E}[\hat{z}_{it}^q | \hat{z}_{i,t-1}^q]$ the true conditional mean. ξ^q enters $\tilde{\xi}$ transformed (heteroskedastic loading, skewed distribution, shock-dependent coefficients).

Why β is identified. Z_{it} is $\sigma(\hat{z}_{i,t-1}^q)$ -measurable (lagged inputs $\Rightarrow$ functions of past z^q). So $\mathbb{E}[\tilde{\xi} \cdot Z] = 0$ iff the *misspec gap* is uncorrelated with Z . Walking through the three DGPs from the previous slide:

- ▶ **(1) AR(1), Gaussian:** $\mu(z_{t-1}^q) = \rho z_{t-1}^q$ is linear; a linear $\tilde{g}$ matches exactly $\Rightarrow$ gap = 0, $\tilde{\xi} = \xi^q$. β exactly identified.
- ▶ **(2) Quadratic-mean location-scale:** $\mu(z_{t-1}^q) = \rho_0 z_{t-1}^q + \rho_1 (z_{t-1}^q)^2$ is quadratic. A quadratic $\tilde{g}$ absorbs the drift $\Rightarrow$ gap = 0. The heteroskedastic loading $(1 + \gamma z_{t-1}^q)\xi^q$ has zero conditional mean and only inflates higher moments of $\tilde{\xi}$ (efficiency loss). β identified once $\tilde{g}$ is quadratic.
- ▶ **(3) ABB polynomial:** $\mu(\cdot)$ remains polynomial in z_{t-1}^q (coefficients = expectations over ξ^q). A polynomial $\tilde{g}$ of matching order absorbs μ asymptotically; shock-dependent coefficients show up only as richer heteroskedasticity in $\tilde{\xi}$ (efficiency cost). β identified asymptotically; finite-sample bias if $\tilde{g}$ is under-parameterized.

In a nutshell: the moment condition tolerates everything that lives in higher moments of $\tilde{\xi}$. The only real threat is misspecifying the conditional mean — fixed by polynomial flexibility.

Heterogeneity: Age

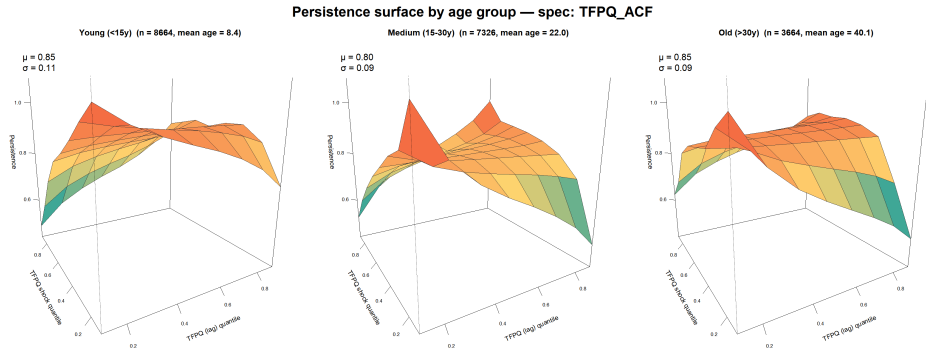


Figure 11: TFPQ persistence by age

Heterogeneity: Size

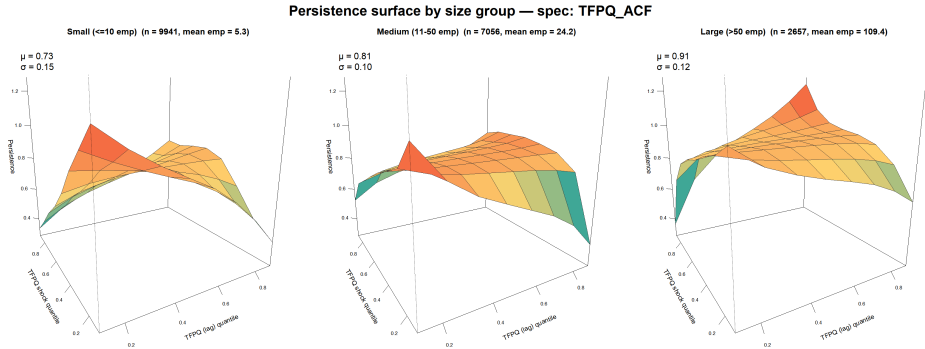


Figure 12: TFPQ persistence by size

Markup Estimation

Production-based approach (De Loecker & Warzynski, 2012):

From cost minimization FOC:

$$\mu_{ist} = \theta_s^M \cdot \frac{P_{ist} Q_{ist}}{P_{ist}^M M_{ist}}$$

- ▶ θ_s^M : materials output elasticity (from production function)
- ▶ Numerator: firm revenue
- ▶ Denominator: materials expenditure

Log markup:

$$\mu_{ist} = \log \theta_s^M + (p + q) - (p^M + m)$$

Marginal cost:

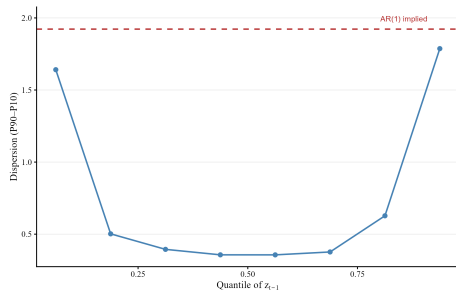
$$mc_{ist} = p_{ist} - \mu_{ist}$$

Convergence of Production Function Estimates

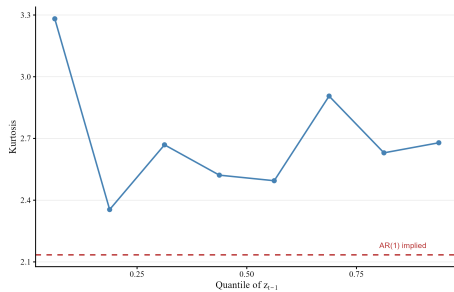
Iteration 0				Iteration 1				Final			
	β^L	β^K	β^M		β^L	β^K	β^M		β^L	β^K	β^M
Bread	0.27	0.04	0.32	Bread	0.14	0.06	0.21	Bread	0.15	0.05	0.20
Metal	0.16	0.05	0.66	Metal	0.13	0.03	0.55	Metal	0.12	0.04	0.60
Moulds	0.26	0.05	0.44	Moulds	0.26	0.05	0.43	Moulds	0.26	0.05	0.44
Footwear	0.40	0.08	0.60	Footwear	0.31	0.04	0.34	Footwear	0.31	0.02	0.36

Conditional Moments of TFPR

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Conditional dispersion



Conditional kurtosis

- ▶ U-shaped dispersion (like TFPQ)
- ▶ Uniformly high kurtosis (~ 2.5 – 2.9): tail risk everywhere
- ▶ Large revenue shocks can strike any firm

TFPQ, TFPR and prices in comparison

Cross-section dispersion: TFPQ, log price, TFPR

Pooled, $n = 24301$ firm-years

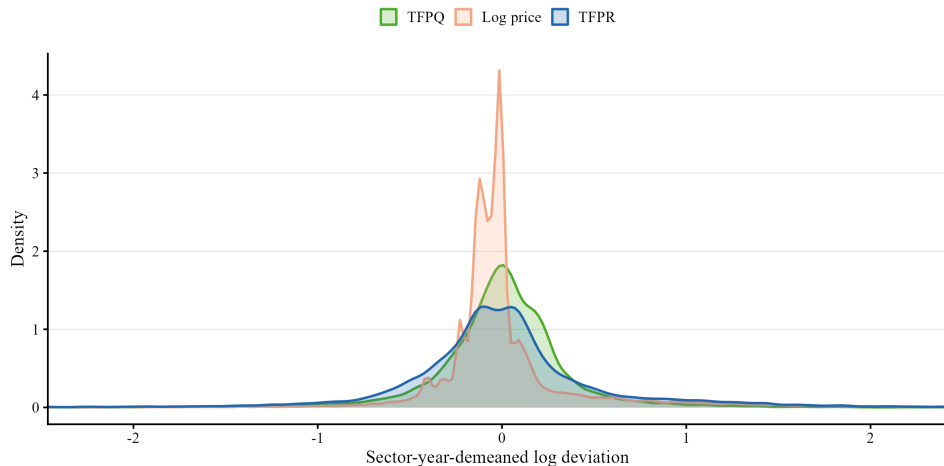


Figure 13: Distributions of TFPQ, TFPR, and prices compared

TFPQ as instrument: $\text{corr}(\omega_{jt}, \log p_{jt})$ per sector

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Sector	N	Iter 1		Final iter	
		filt.	raw	filt.	raw
2512105	8,882	-0.066	-0.268	-0.087	-0.269
1520135	6,166	-0.000	-0.232	-0.010	-0.221
2573507	3,329	-0.282	+0.038	-0.226	+0.038
3102100	1,918	-0.311	-0.724	+0.048	-0.753
1071110	1,828	-0.089	-0.627	-0.061	-0.599
1414300	993	-0.061	-0.358	-0.058	-0.308
1391191	729	-0.139	-0.214	-0.131	-0.214
1051405	702	-0.178	-0.348	-0.163	-0.348

filt. = $\hat{\Phi} - \beta X$ (first-stage filtered ω). **raw** = $y - \beta X$ (no first-stage filter)

References I